Chapter 2

The Supersymmetry Algebra

The last chapter dealt with "why"; this one deals with "what". We want to answer a very concrete question: if we add a pair of fermionic generators Q_α and $\bar{Q}_{\dot{\alpha}}$ to the Poincaré algebra, what must their (anti)commutation relations with P_μ and $L_{\mu\nu}$, and with each other, look like?

You will see that almost every answer is *forced* on us. Lorentz covariance first restricts what can possibly appear on the right-hand side of each relation, and then the Jacobi identities pin down the remaining coefficients one by one. In the end only one coefficient is left undetermined, and it can be absorbed into a normalization. Every tool the derivation uses was already prepared in Chapter 0.

2.1 Fermionic Generators

The supersymmetry generator Q turns bosons into fermions and fermions into bosons:

$$Q|B\rangle = |F\rangle, \qquad Q|F\rangle = |B'\rangle. \tag{2.1.1}$$

Bosons have integer spin and fermions half-integer spin, so Q must change the angular momentum by half a unit. That means Q itself carries half-integer spin: it transforms as a spinor under Lorentz transformations. The simplest choice is a left-handed Weyl spinor $Q_\alpha \in (\frac{1}{2}, 0)$. Its Hermitian conjugate is written

$$\bar{Q}_{\dot{\alpha}} \equiv (Q_\alpha)^\dagger, \tag{2.1.2}$$

and by the rule of Section 0.2.2, complex conjugation turns (j_L, j_R) into (j_R, j_L), so $\bar{Q}_{\dot{\alpha}} \in (0, \frac{1}{2})$.

2.1.1 Boson or Fermion: $(-1)^F$

Introduce the operator $(-1)^F$, which is $+1$ on bosonic states and -1 on fermionic states:

$$(-1)^F|B\rangle = +|B\rangle, \qquad (-1)^F|F\rangle = -|F\rangle. \tag{2.1.3}$$

(F is the fermion number, so $(-1)^F$ only cares whether the number of fermions is even or odd.) Q turns bosonic states into fermionic ones and vice versa, so it *anti*commutes with $(-1)^F$:

$$\boxed{\{(-1)^F, Q_\alpha\} = 0.} \tag{2.1.4}$$

You can check this on a bosonic state: $(-1)^F Q|B\rangle = -Q|B\rangle$, while $Q(-1)^F|B\rangle = +Q|B\rangle$, and the two add to zero. We say Q is "odd", while ordinary generators like P_μ and $L_{\mu\nu}$ are "even".

2.1.2 Graded Lie Algebras

The product of two odd operators is even. So the relation between two Q's should have even generators on the right. The question is: should we use the commutator $[Q, Q']$ or the

anticommutator $\{Q, Q'\}$? Both are even. The answer comes from where Q comes from. Like P_μ, Q is the spatial integral of some conserved current, and that current is built from fermion fields. Fermion fields satisfy anticommutation relations (recall Section 0.6.1), so the natural operation between two fermionic generators is the anticommutator. This falls squarely outside the fourth assumption of the Coleman–Mandula theorem: it is exactly the loophole of the last chapter.

In general, give each generator J_a a "grading" η_a: $\eta_a = 0$ for even generators and $\eta_a = 1$ for odd ones. Define the **superbracket**

$$\boxed{[J_a, J_b\} \equiv J_a J_b - (-1)^{\eta_a \eta_b} J_b J_a = \mathrm{i} f_{ab}{}^c J_c.} \tag{2.1.5}$$

If at least one of the two is even, this is the ordinary commutator; if both are odd, it is the anticommutator. Only generators whose grading "adds up" can appear on the right: even with even gives even, even with odd gives odd, odd with odd gives even. An algebra obeying relations of this kind is called a **graded Lie algebra**, or a **superalgebra**.

Ordinary Lie algebras have the Jacobi identity, and so do superalgebras, except that the signs follow the grading. These are the ones we will use (B stands for an even generator, F for an odd one):

$$\begin{aligned}
[[B_1, B_2], B_3] + [[B_2, B_3], B_1] + [[B_3, B_1], B_2] &= 0, \\
[[B_1, B_2], F] + [[B_2, F], B_1] + [[F, B_1], B_2] &= 0, \\
[\{F_1, F_2\}, B] + \{[B, F_1], F_2\} + \{[B, F_2], F_1\} &= 0, \\
[\{F_1, F_2\}, F_3] + [\{F_2, F_3\}, F_1] + [\{F_3, F_1\}, F_2] &= 0.
\end{aligned} \tag{2.1.6}$$

These require no assumptions: expand all the brackets and the terms cancel in pairs (Problem 2.1). So they hold automatically for any operators on Hilbert space. That is precisely why they can be used to constrain the algebra: if we write some arbitrary right-hand side for a bracket like $[Q, P]$, these identities need not hold any more once we substitute it in, yet they must hold. This is the weapon we will use again and again below.

2.2 Deriving the $\mathcal{N} = 1$ Supersymmetry Algebra

The generators we now have are: P_μ and $L_{\mu\nu}$ of the Poincaré algebra (both Hermitian), whatever internal symmetry generators T_a there may be, and in addition one Q_α and its conjugate $\bar{Q}_{\dot{\alpha}}$. The case with a single Q_α is called $\mathcal{N} = 1$ supersymmetry. We already know the Poincaré algebra itself, and five more relations remain to be determined:

$$[Q_\alpha, L_{\mu\nu}], \qquad [Q_\alpha, P_\mu], \qquad \{Q_\alpha, Q_\beta\}, \qquad \{Q_\alpha, \bar{Q}_{\dot{\beta}}\}, \qquad [Q_\alpha, T_a]. \tag{2.2.1}$$

The next five subsections deal with them in turn. (The remaining relations involving $\bar{Q}$ follow by Hermitian conjugation.)

The strategy has two steps. Step one is Lorentz covariance: the left side of each bracket is a product of two objects, transforming as the tensor product of their two representations; only generators that transform the same way, and whose grading matches, can appear on the right. There are only a few kinds of generators to choose from:

$$P_\mu \in \left(\tfrac{1}{2}, \tfrac{1}{2}\right), \quad L_{\mu\nu} \in (1,0) \oplus (0,1), \quad Q_\alpha \in \left(\tfrac{1}{2}, 0\right), \quad \bar{Q}^{\dot{\alpha}} \in \left(0, \tfrac{1}{2}\right), \quad T_a \in (0,0). \tag{2.2.2}$$

Here $L_{\mu\nu}$ is handled exactly like the field strength in Section 0.7.3: any antisymmetric tensor can be split into self-dual and anti-self-dual halves according to Eq. (0.7.11),

$$L_{\mu\nu} = \frac{1}{2}(\sigma_{\mu\nu})_\alpha{}^\beta L_\beta{}^\alpha + \frac{1}{2}(\bar\sigma_{\mu\nu})^{\dot\alpha}{}_{\dot\beta}\,\bar L^{\dot\beta}{}_{\dot\alpha}, \tag{2.2.3}$$

so its six components split into two sets of three, belonging to $(1,0)$ and $(0,1)$ respectively. Step two is to fix the coefficients using the Jacobi identities (2.1.6).

2.2.1 Step One: Q Is a Spinor

What exactly does it mean to say that "Q_α is a left-handed Weyl spinor"? Q_α is an operator acting on the space of states. Under a Lorentz transformation, states transform with a unitary operator,

$$|\psi\rangle \to U|\psi\rangle, \qquad U = \exp\Big(-\frac{\mathrm{i}}{2}\omega_{\mu\nu}L^{\mu\nu}\Big), \tag{2.2.4}$$

where $L^{\mu\nu}$ are the Hermitian generators on the space of states. The expectation value of an operator then becomes

$$\langle\psi'|Q_\alpha|\psi'\rangle = \langle\psi|U^\dagger Q_\alpha U|\psi\rangle. \tag{2.2.5}$$

Transforming the states and transforming the operators have the same effect (just like the relation between the Schrödinger and Heisenberg pictures in quantum mechanics), so we can say that under Lorentz transformations $Q_\alpha \to U^\dagger Q_\alpha U$. "$Q_\alpha$ is a left-handed spinor" means that the operator transformed this way is exactly what you get by mixing the two components of Q_α with the $(\frac{1}{2},0)$ matrix:

$$U^\dagger Q_\alpha U = M_\alpha{}^\beta Q_\beta, \qquad M = \exp\Big(-\frac{\mathrm{i}}{2}\omega_{\mu\nu}\sigma^{\mu\nu}\Big), \tag{2.2.6}$$

where $\sigma^{\mu\nu}$ are the left-handed generators of Eq. (0.5.26) (the upper block in Section 0.5.4). Whatever you do, don't drop the i in the exponent: without it, M is not an element of $\mathrm{SL}(2,\mathbb{C})$.

Take ω infinitesimal and expand both sides to first order. The left side is

$$U^\dagger Q_\alpha U = \Big(1+\frac{\mathrm{i}}{2}\omega_{\mu\nu}L^{\mu\nu}\Big)Q_\alpha\Big(1-\frac{\mathrm{i}}{2}\omega_{\mu\nu}L^{\mu\nu}\Big) = Q_\alpha + \frac{\mathrm{i}}{2}\omega_{\mu\nu}\,[L^{\mu\nu},Q_\alpha] + O(\omega^2), \tag{2.2.7}$$

and the right side is $Q_\alpha - \frac{\mathrm{i}}{2}\omega_{\mu\nu}(\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta + O(\omega^2)$. The $\omega_{\mu\nu}$ are arbitrary antisymmetric parameters, and what multiplies them on both sides is antisymmetric in $\mu\nu$, so the coefficients must match term by term:

$$[L^{\mu\nu},Q_\alpha] = -(\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta, \qquad \text{that is,} \qquad \boxed{[Q_\alpha,L^{\mu\nu}] = (\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta.} \tag{2.2.8}$$

Doing the same for $\bar Q^{\dot\alpha}$ (or taking the Hermitian conjugate of the result above) gives

$$\boxed{[\bar Q^{\dot\alpha},L^{\mu\nu}] = (\bar\sigma^{\mu\nu})^{\dot\alpha}{}_{\dot\beta}\,\bar Q^{\dot\beta}.} \tag{2.2.9}$$

Is that minus sign right? There is an independent check: substitute Eq. (2.2.8) into the Jacobi identity $[[L,L],Q]+\cdots=0$, and it requires the matrices $\sigma^{\mu\nu}$ to obey exactly the same commutation relations (0.1.33) as $L^{\mu\nu}$. Since $\sigma^{\mu\nu}$ is a representation of the Lorentz algebra to begin with, the check passes (Problem 2.2).

From the point of view of representation theory, this result is in fact the only possibility. $[Q_\alpha, L_{\mu\nu}]$ is odd and transforms as

$$\Big(\tfrac{1}{2},0\Big)\otimes\big[(1,0)\oplus(0,1)\big] = \Big(\tfrac{3}{2},0\Big)\oplus\Big(\tfrac{1}{2},0\Big)\oplus\Big(\tfrac{1}{2},1\Big). \tag{2.2.10}$$

(Looking only at the left-handed copy of angular momentum, this is $\frac{1}{2}\otimes 1 = \frac{3}{2}\oplus\frac{1}{2}$, with dimensions $2\times 3 = 4+2$.) There are no odd generators in $(\frac{3}{2},0)$ or $(\frac{1}{2},1)$; only Q_α itself belongs to $(\frac{1}{2},0)$. So the right side can only be proportional to $\sigma^{\mu\nu}Q$, and the constant of proportionality is fixed by the requirement that Q be a spinor. While we're at it, a reminder of who looks after which index: the undotted index α belongs to the left-handed copy of angular momentum J^L, and the dotted index $\dot\alpha$ to J^R.

2.2.2 Step Two: Q Commutes with Translations

$[Q_\alpha, P_\mu]$ is odd and transforms as

$$\left(\tfrac{1}{2},0\right)\otimes\left(\tfrac{1}{2},\tfrac{1}{2}\right) = \left(1,\tfrac{1}{2}\right)\oplus\left(0,\tfrac{1}{2}\right). \tag{2.2.11}$$

There are no generators in $(1,\frac{1}{2})$, and the only one in $(0,\frac{1}{2})$ is $\bar Q^{\dot\alpha}$. To "translate" a vector index and a dotted index into an undotted index we use the invariant tensor σ_μ (Section 0.5.1), so the most general form is

$$[Q_\alpha, P_\mu] = c\,(\sigma_\mu)_{\alpha\dot\alpha}\bar Q^{\dot\alpha}, \tag{2.2.12}$$

where c is a complex number to be determined. Taking the Hermitian conjugate gives (Problem 2.3)

$$[\bar Q^{\dot\alpha}, P_\mu] = c^*(\bar\sigma_\mu)^{\dot\alpha\alpha}Q_\alpha. \tag{2.2.13}$$

Now use a Jacobi identity (the second line of Eq. (2.1.6), with $B_1 = P_\mu$, $B_2 = P_\nu$, $F = Q_\alpha$):

$$[P_\mu,[P_\nu,Q_\alpha]] + [P_\nu,[Q_\alpha,P_\mu]] + [Q_\alpha,[P_\mu,P_\nu]] = 0. \tag{2.2.14}$$

The last term vanishes, since translations commute with each other. Compute the first two one at a time:

$$\begin{aligned}
[P_\mu,[P_\nu,Q_\alpha]] &= -c\,(\sigma_\nu)_{\alpha\dot\alpha}[P_\mu,\bar Q^{\dot\alpha}] = |c|^2(\sigma_\nu\bar\sigma_\mu)_\alpha{}^\beta Q_\beta,\\
[P_\nu,[Q_\alpha,P_\mu]] &= c\,(\sigma_\mu)_{\alpha\dot\alpha}[P_\nu,\bar Q^{\dot\alpha}] = -|c|^2(\sigma_\mu\bar\sigma_\nu)_\alpha{}^\beta Q_\beta.
\end{aligned} \tag{2.2.15}$$

Adding them,

$$|c|^2\big(\sigma_\nu\bar\sigma_\mu - \sigma_\mu\bar\sigma_\nu\big)_\alpha{}^\beta Q_\beta = 4\mathrm{i}\,|c|^2(\sigma_{\mu\nu})_\alpha{}^\beta Q_\beta = 0, \tag{2.2.16}$$

where the middle step uses the definition (0.5.26) of $\sigma^{\mu\nu}$. $\sigma_{\mu\nu}$ is not the zero matrix, and Q_1, Q_2 are two independent operators, so we must have $|c|^2 = 0$:

$$\boxed{[Q_\alpha,P_\mu]=0, \qquad [\bar Q_{\dot\alpha},P_\mu]=0.} \tag{2.2.17}$$

The physical meaning of this result is direct: a supersymmetry transformation doesn't change momentum. A state of momentum p, acted on by Q, still has momentum p. In the next section we will see that this immediately implies that superpartners have the same mass.

2.2.3 Step Three: Two Q's Anticommute

$\{Q_\alpha, Q_\beta\}$ is even and transforms as

$$\left(\tfrac{1}{2},0\right)\otimes\left(\tfrac{1}{2},0\right) = (1,0)\oplus(0,0). \tag{2.2.18}$$

There is a detail here worth going through slowly. In Section 0.6.2 we saw that a combination of two spinor indices splits into an antisymmetric part (proportional to $\varepsilon_{\alpha\beta}$, belonging to $(0,0)$)

and a symmetric part (belonging to $(1,0)$). And the anticommutator $\{Q_\alpha, Q_\beta\}$ is obviously unchanged by swapping $\alpha \leftrightarrow \beta$: it is *symmetric*. So the $(0,0)$ part simply cannot appear. Notice that the reason here is symmetry, not "there are no even generators in $(0,0)$": the internal symmetry generators T_a are $(0,0)$, but they could only appear in the form $\varepsilon_{\alpha\beta}T_a$, which is antisymmetric. (In Section 2.4, when there is more than one Q, this door will open again.)

For the remaining $(1,0)$ part, the only even generator that qualifies is the self-dual half of $L_{\mu\nu}$. Therefore

$$\{Q_\alpha, Q^\beta\} = c' \, (\sigma^{\mu\nu})_\alpha{}^\beta L_{\mu\nu}, \tag{2.2.19}$$

where we have raised the index of Q_β so that its position matches the indices of $(\sigma^{\mu\nu})_\alpha{}^\beta$. Now use the third line of Eq. (2.1.6), with $F_1 = Q_\alpha$, $F_2 = Q^\beta$, $B = P_\rho$:

$$[\{Q_\alpha, Q^\beta\}, P_\rho] + \{[P_\rho, Q_\alpha], Q^\beta\} + \{[P_\rho, Q^\beta], Q_\alpha\} = 0. \tag{2.2.20}$$

By Eq. (2.2.17), the last two terms vanish. Compute the first using Eq. (0.1.29):

$$c'(\sigma^{\mu\nu})_\alpha{}^\beta [L_{\mu\nu}, P_\rho] = \mathrm{i}c'(\sigma^{\mu\nu})_\alpha{}^\beta \big(\eta_{\nu\rho}P_\mu - \eta_{\mu\rho}P_\nu\big) = 2\mathrm{i}c'(\sigma^\mu{}_\rho)_\alpha{}^\beta P_\mu. \tag{2.2.21}$$

This must vanish for every momentum, and $\sigma^\mu{}_\rho$ is not the zero matrix, so $c' = 0$:

$$\boxed{\{Q_\alpha, Q_\beta\} = 0, \qquad \{\bar{Q}_{\dot\alpha}, \bar{Q}_{\dot\beta}\} = 0.} \tag{2.2.22}$$

In particular, $(Q_1)^2 = (Q_2)^2 = 0$: the same Q acting twice in a row gives zero. This is just like the Pauli exclusion principle, and it is what will make supermultiplets contain only finitely many states.

2.2.4 Step Four: Q and $\bar{Q}$

$\{Q_\alpha, \bar{Q}_{\dot\beta}\}$ is even and transforms as $(\frac{1}{2},0) \otimes (0,\frac{1}{2}) = (\frac{1}{2},\frac{1}{2})$. The only even generator in $(\frac{1}{2},\frac{1}{2})$ is P_μ, so

$$\{Q_\alpha, \bar{Q}_{\dot\beta}\} = c''(\sigma^\mu)_{\alpha\dot\beta}P_\mu. \tag{2.2.23}$$

This time the Jacobi identities are no help: they hold automatically whatever c'' is (Problem 2.4). Still, c'' is very nearly pinned down anyway, by the following three arguments.

First, c'' cannot be zero. Take $\dot\beta$ to have the same numerical value as α, so that $\bar{Q}_{\dot\beta} = Q_\alpha^\dagger$. If $c'' = 0$, then for any state $|\psi\rangle$,

$$0 = \langle\psi|\{Q_\alpha, Q_\alpha^\dagger\}|\psi\rangle = \big\|Q_\alpha^\dagger|\psi\rangle\big\|^2 + \big\|Q_\alpha|\psi\rangle\big\|^2. \tag{2.2.24}$$

Two nonnegative numbers adding to zero must each be zero, so Q_α sends every state to zero: there is no supersymmetry at all. So $\{Q, \bar{Q}\} \neq 0$ is the very heart of the algebra.

Second, the size of c'' is a convention. Replace Q by κQ ($\kappa > 0$) and c'' becomes $\kappa^2 c''$, so we can make $|c''|$ whatever we like.

Third, c'' must be a positive real number. By the same norm argument, $\{Q_\alpha, Q_\alpha^\dagger\}$ is a positive operator: its expectation value in any state is a sum of two squared norms; and for a one-particle state of positive energy, the eigenvalues of $\sigma^\mu p_\mu = p_0\mathbf{1} + \sigma_i p_i$ are $p^0 \pm |\boldsymbol{p}| \geq 0$. For the two sides to agree, c'' can only be a positive real number.

By convention we take $c'' = 2$:

$$\boxed{\{Q_\alpha, \bar{Q}_{\dot\beta}\} = 2(\sigma^\mu)_{\alpha\dot\beta}P_\mu.} \tag{2.2.25}$$

This is the single most important formula in the supersymmetry algebra. It says: do two supersymmetry transformations in a row, and the result is a translation. That is why people often say supersymmetry is "the square root of a spacetime translation". Supersymmetry is not an "internal" symmetry; it is entangled with spacetime itself.

2.2.5 Step Five: Q and Internal Symmetries

Finally, look at $[Q_\alpha, T_a]$. It is odd and belongs to $(\frac{1}{2}, 0)$; in the $\mathcal{N} = 1$ case the only such generator is Q_α itself, so

$$[T_a, Q_\alpha] = b_a\, Q_\alpha, \tag{2.2.26}$$

where the b_a are some numbers. Taking the Hermitian conjugate gives $[T_a, \bar{Q}_{\dot\alpha}] = -b_a^* \bar{Q}_{\dot\alpha}$. Now use a Jacobi identity (the third line of Eq. (2.1.6), with $B = T_a$):

$$[T_a, \{Q_\alpha, \bar{Q}_{\dot\beta}\}] = \{[T_a, Q_\alpha], \bar{Q}_{\dot\beta}\} + \{Q_\alpha, [T_a, \bar{Q}_{\dot\beta}]\} = (b_a - b_a^*)\{Q_\alpha, \bar{Q}_{\dot\beta}\}. \tag{2.2.27}$$

The left side is $2\sigma^\mu[T_a, P_\mu] = 0$ (internal symmetries commute with translations), so b_a is real. Moreover, the b_a must furnish a one-dimensional representation of the internal symmetry algebra: $[[T_a, T_b], Q] = [T_a, [T_b, Q]] - [T_b, [T_a, Q]] = 0$, so $f_{ab}{}^c b_c = 0$. For the non-abelian (semisimple) part, this forces $b_a = 0$.

The conclusion: Q_α commutes with all the generators of internal symmetries,

$$[Q_\alpha, T_a] = 0, \tag{2.2.28}$$

with *one exception* at most: a single abelian $U(1)$ generator, which can multiply Q by a phase. That is the R-symmetry of the next section. (Incidentally, $(-1)^F$ is not one of the T_a here: it is a discrete operator, and by Eq. (2.1.4) it anticommutes with Q rather than commuting with it.)

2.2.6 The Result

Putting the conclusions of this section together with the Poincaré algebra, we obtain the $\mathcal{N} = 1$ **super-Poincaré algebra**:

$$\boxed{\begin{aligned}
&[P_\mu, P_\nu] = 0, \qquad [L_{\mu\nu}, P_\rho] = \mathrm{i}(\eta_{\nu\rho}P_\mu - \eta_{\mu\rho}P_\nu), \qquad [L, L] \text{ as in Eq. (0.1.33)},\\
&[Q_\alpha, P_\mu] = [\bar{Q}_{\dot\alpha}, P_\mu] = 0,\\
&[Q_\alpha, L^{\mu\nu}] = (\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta, \qquad [\bar{Q}^{\dot\alpha}, L^{\mu\nu}] = (\bar\sigma^{\mu\nu})^{\dot\alpha}{}_{\dot\beta}\bar{Q}^{\dot\beta},\\
&\{Q_\alpha, Q_\beta\} = \{\bar{Q}_{\dot\alpha}, \bar{Q}_{\dot\beta}\} = 0, \qquad \{Q_\alpha, \bar{Q}_{\dot\beta}\} = 2(\sigma^\mu)_{\alpha\dot\beta}P_\mu.
\end{aligned}} \tag{2.2.29}$$

Looking back, we made almost no "choices": apart from the normalization of c'', everything was forced on us by Lorentz covariance and the Jacobi identities.

Example 2.1 Energy is a sum of squares of supercharges

In Eq. (2.2.25), take $\dot\beta$ to have the same numerical value as α, and sum over $\alpha = 1, 2$. Show that

$$H \equiv P^0 = \frac{1}{4}\sum_{\alpha=1,2}\Big(Q_\alpha Q_\alpha^\dagger + Q_\alpha^\dagger Q_\alpha\Big). \tag{2.2.30}$$

Solution The left side is $\sum_\alpha\{Q_\alpha, \bar{Q}_{\dot\alpha}\}$ (with $\dot\alpha$ taking the same numerical value as α), and since $\bar{Q}_{\dot\alpha} = Q_\alpha^\dagger$, it is exactly the sum in parentheses. The right side is $2\sum_\alpha(\sigma^\mu)_{\alpha\alpha}P_\mu = 2\,\mathrm{tr}(\sigma^\mu)P_\mu$.

The trace of $\sigma^0 = \mathbf{1}$ is 2, and the three Pauli matrices are traceless, so the right side equals $4P_0 = 4P^0 = 4H$. Divide both sides by 4 and you're done. Short as it is, this formula will lead to a rather startling conclusion in Section 2.5. □

2.3 R-Symmetry

An **automorphism** of an algebra is a recombination of the generators that leaves all the bracket relations unchanged. Take a careful look at Eq. (2.2.29): Q pairs with $\bar{Q}$ only in the form $\{Q, \bar{Q}\}$, and every other relation is either linear or simply zero. So if we multiply Q by a phase and simultaneously multiply $\bar{Q}$ by the opposite phase,

$$Q_\alpha \to e^{\mathrm{i}\theta} Q_\alpha, \qquad \bar{Q}_{\dot\alpha} \to e^{-\mathrm{i}\theta} \bar{Q}_{\dot\alpha}, \qquad \theta \in \mathbb{R}, \tag{2.3.1}$$

the whole algebra doesn't budge (θ is real, so $\bar{Q}$ is still the Hermitian conjugate of Q). These phase transformations form a group, $U(1)_R$.

If a theory actually has this symmetry, there is a Hermitian generator R whose unitary operator $U_R = e^{-\mathrm{i}\theta R}$ satisfies $U_R^\dagger Q_\alpha U_R = e^{\mathrm{i}\theta} Q_\alpha$. Expanding to first order just as in Section 2.2.1:

$$Q_\alpha + \mathrm{i}\theta [R, Q_\alpha] = Q_\alpha + \mathrm{i}\theta Q_\alpha \qquad \Longrightarrow \qquad \boxed{[R, Q_\alpha] = Q_\alpha, \qquad [R, \bar{Q}_{\dot\alpha}] = -\bar{Q}_{\dot\alpha}.} \tag{2.3.2}$$

R is an internal symmetry, so it commutes with both P_μ and $L_{\mu\nu}$.

This formula has an intuitive meaning. Let $|r\rangle$ be an eigenstate of R with eigenvalue (R-charge) r. Then

$$R\, Q_\alpha |r\rangle = \big([R, Q_\alpha] + Q_\alpha R\big)|r\rangle = (r+1)\, Q_\alpha |r\rangle. \tag{2.3.3}$$

Q raises the R-charge by 1, and $\bar{Q}$ lowers it by 1. So within one supermultiplet, a boson and its fermionic partner have R-charges differing by 1. That is exactly how R-symmetry differs from ordinary internal symmetries: an ordinary internal symmetry treats all members of a supermultiplet alike, so it commutes with Q; R-symmetry treats them differently, so it does not commute with Q. It is the one exception the Haag–Łopuszański–Sohnius theorem allows.

Note that the algebra only *permits* this symmetry; whether a particular theory has it depends on the Lagrangian. When we construct Lagrangians later, we will see that some interactions preserve R-symmetry while others break it.

2.4 Extended Supersymmetry and Central Charges

Nothing says there can only be one Q. Suppose there are $\mathcal{N}$ copies of the supercharges,

$$Q^I_\alpha, \qquad \bar{Q}_{\dot\alpha I} \equiv \big(Q^I_\alpha\big)^\dagger, \qquad I = 1, \ldots, \mathcal{N}. \tag{2.4.1}$$

This is called **extended supersymmetry**. The five-step derivation of the last section can be repeated almost word for word, except that each relation now carries extra I, J indices. Step one is unchanged: each Q^I is a left-handed spinor. Step two still gives $[Q^I_\alpha, P_\mu] = 0$. Step four gives $\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\} = 2(\sigma^\mu)_{\alpha\dot\beta} P_\mu\, \delta^I{}_J$, where we have chosen a suitable basis for the Q^I to turn the coefficient matrix into the identity.

What really changes is step three. Now the anticommutator $\{Q^I_\alpha, Q^J_\beta\}$ is unchanged when we swap $(\alpha, I) \leftrightarrow (\beta, J)$ *simultaneously*. There are two ways to achieve this: symmetric in $\alpha\beta$ and

symmetric in IJ, or antisymmetric in $\alpha\beta$ and antisymmetric in IJ. The first belongs to $(1,0)$ and is wiped out by the Jacobi identities just as in Section 2.2.3. The second belongs to $(0,0)$ and is proportional to $\varepsilon_{\alpha\beta}$, and this time it is allowed:

$$\boxed{\{Q^I_\alpha, Q^J_\beta\} = \varepsilon_{\alpha\beta} Z^{IJ}, \qquad Z^{IJ} = -Z^{JI},} \tag{2.4.2}$$

and taking the Hermitian conjugate, $\{\bar{Q}_{\dot{\alpha}I}, \bar{Q}_{\dot{\beta}J}\} = \varepsilon_{\dot{\alpha}\dot{\beta}} \bar{Z}_{IJ}$, with $\bar{Z}_{IJ} \equiv (Z^{IJ})^\dagger$. For $\mathcal{N} = 1$, $Z^{11} = -Z^{11} = 0$, consistent with the conclusion of Section 2.2.3.

What is Z^{IJ}? It is an even, Lorentz-scalar generator. A Jacobi identity (the third line of Eq. (2.1.6), with $B = P_\mu$) immediately gives $\varepsilon_{\alpha\beta}[Z^{IJ}, P_\mu] = 0$, so Z commutes with translations; it is a Lorentz scalar, so it commutes with $L_{\mu\nu}$. A further analysis (part of the work of Haag, Łopuszański and Sohnius) shows that Z^{IJ} commutes with *every* generator of the superalgebra. Generators that commute with everything form the **center** of the algebra, so the Z^{IJ} are called **central charges**. Z^{IJ} is an $\mathcal{N} \times \mathcal{N}$ antisymmetric matrix, giving $\mathcal{N}(\mathcal{N}-1)/2$ independent complex central charges in all. We will see later that central charges set a lower bound on the masses of particles.

The automorphisms of the extended superalgebra are larger too. Take any $\mathcal{N} \times \mathcal{N}$ unitary matrix $\mathcal{R}$ and let

$$Q^I_\alpha \to \mathcal{R}^I{}_J Q^J_\alpha, \qquad \bar{Q}_{\dot{\alpha}I} \to \bar{Q}_{\dot{\alpha}J} (\mathcal{R}^\dagger)^J{}_I, \qquad \mathcal{R}\mathcal{R}^\dagger = \mathbf{1}. \tag{2.4.3}$$

Unitarity guarantees that the $\delta^I{}_J$ on the right of $\{Q, \bar{Q}\}$ is unchanged, so (when the central charges vanish) the algebra stays the same. Such transformations form $U(\mathcal{N})_R$, the generalization of the $U(1)_R$ of Section 2.3. If the central charges are nonzero, only the subgroup that leaves Z^{IJ} invariant is an automorphism.

How large can $\mathcal{N}$ be? Each Q^I_α has two complex components, that is, four real components, so $\mathcal{N}$ copies of the supercharges have $4\mathcal{N}$ real components in all. We will see when we construct supermultiplets that without gravity, and requiring helicities no larger than 1, we need $\mathcal{N} \leq 4$; with gravity (helicities up to 2), $\mathcal{N} \leq 8$. The $\mathcal{N} = 4$ SYM mentioned in the last chapter is precisely the gauge theory with the most supersymmetry possible without gravity.

2.5 First Steps in Representation Theory

In Section 0.3.2 we labeled particles with the two Casimir operators of the Poincaré algebra, P^2 and W^2. An irreducible representation of the super-Poincaré algebra is called a **supermultiplet**; it contains several Poincaré representations (several kinds of particle), related to each other by Q. Before we start constructing supermultiplets, let us prove three properties that hold for all of them.

2.5.1 Same Mass, Different Spins

By Eq. (2.2.17), Q commutes with P_μ, and hence with P^2:

$$[P^2, Q_\alpha] = 0. \tag{2.5.1}$$

So P^2 is still a Casimir operator: *all the particles in a supermultiplet have the same mass.* If supersymmetry were unbroken, the electron would have a scalar partner of exactly the same mass.

We don't see such a particle, and that is the experimental basis for the statement "supersymmetry must be broken".

Not so for W^2. Q doesn't commute with $L_{\mu\nu}$, so it doesn't commute with W^μ either:

$$[W^\mu, Q_\alpha] = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma} P_\nu [L_{\rho\sigma}, Q_\alpha] = -P_\nu \Big(\frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}\sigma_{\rho\sigma}\Big)_\alpha{}^\beta Q_\beta = \mathrm{i} P_\nu (\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta. \tag{2.5.2}$$

The last step uses the self-duality condition (0.6.12). This is just what we expected: Q changes spin, the particles in one supermultiplet have different spins, and of course W^2 can't be a constant.

Can we patch up W^μ to build a new Casimir that commutes with Q? Let's try adding to W^μ a vector built from Q and $\bar{Q}$,

$$X^\mu \equiv \bar{Q}_{\dot\alpha}(\bar\sigma^\mu)^{\dot\alpha\alpha} Q_\alpha. \tag{2.5.3}$$

First compute its commutator with Q_γ. Using $Q_\alpha Q_\gamma = -Q_\gamma Q_\alpha$ (Eq. (2.2.22)),

$$[X^\mu, Q_\gamma] = -\big(\bar{Q}_{\dot\alpha} Q_\gamma + Q_\gamma \bar{Q}_{\dot\alpha}\big)(\bar\sigma^\mu)^{\dot\alpha\alpha} Q_\alpha = -\{Q_\gamma, \bar{Q}_{\dot\alpha}\}(\bar\sigma^\mu)^{\dot\alpha\alpha} Q_\alpha = -2P_\nu (\sigma^\nu \bar\sigma^\mu)_\gamma{}^\alpha Q_\alpha. \tag{2.5.4}$$

From Eqs. (0.2.14) and (0.5.26), $\sigma^\nu\bar\sigma^\mu = \eta^{\mu\nu} + 2\mathrm{i}\sigma^{\mu\nu}$ (Problem 2.7). So for $\tilde{W}^\mu \equiv W^\mu + aX^\mu$,

$$[\tilde{W}^\mu, Q_\alpha] = P_\nu \Big[(\mathrm{i} - 4\mathrm{i}a)\,\sigma^{\mu\nu} - 2a\,\eta^{\mu\nu}\Big]_\alpha{}^\beta Q_\beta. \tag{2.5.5}$$

Taking $a = \frac{1}{4}$ makes the $\sigma^{\mu\nu}$ term disappear, leaving $[\tilde{W}^\mu, Q_\alpha] = -\frac{1}{2}P^\mu Q_\alpha$. What's left is proportional to P^μ, and antisymmetrizing with P gets rid of it:

$$\hat{W}^{\mu\nu} \equiv \tilde{W}^\mu P^\nu - \tilde{W}^\nu P^\mu, \qquad [\hat{W}^{\mu\nu}, Q_\alpha] = -\frac{1}{2}\big(P^\mu P^\nu - P^\nu P^\mu\big) Q_\alpha = 0. \tag{2.5.6}$$

To sum up,

$$\boxed{\tilde{W}^\mu = W^\mu + \frac{1}{4}\bar{Q}_{\dot\alpha}(\bar\sigma^\mu)^{\dot\alpha\alpha} Q_\alpha, \qquad \hat{W}^{\mu\nu} = \tilde{W}^\mu P^\nu - \tilde{W}^\nu P^\mu, \qquad \tilde{C}_2 = \hat{W}_{\mu\nu}\hat{W}^{\mu\nu}.} \tag{2.5.7}$$

$\tilde{C}_2$ commutes with Q, with $\bar{Q}$ (take the Hermitian conjugate), and with P, and it is a Lorentz scalar, so it is the second Casimir operator of the super-Poincaré algebra. Its eigenvalue labels supermultiplets and is called the **superspin**; we will compute its value when we construct massive supermultiplets.[1]

2.5.2 As Many Bosons as Fermions

At the end of Chapter 0 we noticed that on shell the chiral multiplet has exactly two bosonic states and two fermionic states. This is no coincidence; it is a requirement of the algebra: *in any supermultiplet with nonzero momentum, there are as many bosonic states as fermionic states.*

The proof is short. Fix a momentum $p \neq 0$, and consider the space V_p spanned by all states in the supermultiplet with momentum p. It is finite-dimensional, and since Q and $\bar{Q}$ commute with P, they map V_p into itself. Take the trace of $(-1)^F\{Q_\alpha, \bar{Q}_{\dot\beta}\}$ over V_p. On the one hand, using cyclicity of the trace and Eq. (2.1.4),

$$\mathrm{tr}\,\big[(-1)^F Q_\alpha \bar{Q}_{\dot\beta}\big] = \mathrm{tr}\,\big[\bar{Q}_{\dot\beta}(-1)^F Q_\alpha\big] = -\,\mathrm{tr}\,\big[\bar{Q}_{\dot\beta} Q_\alpha (-1)^F\big] = -\,\mathrm{tr}\,\big[(-1)^F \bar{Q}_{\dot\beta} Q_\alpha\big], \tag{2.5.8}$$

[1] With the convention $\varepsilon^{0123} = +1$, W^μ changes sign as a whole, and the coefficient of the correction term becomes $-\frac{1}{4}$ accordingly. That is why you will see $W^\mu - \frac{1}{4}\bar{Q}\bar\sigma^\mu Q$ in other books.

so $\mathrm{tr}[(-1)^F\{Q_\alpha, \bar{Q}_{\dot\beta}\}] = 0$. On the other hand, by Eq. (2.2.25),

$$\mathrm{tr}\left[(-1)^F\{Q_\alpha, \bar{Q}_{\dot\beta}\}\right] = 2(\sigma^\mu)_{\alpha\dot\beta}\, p_\mu\, \mathrm{tr}(-1)^F = 2(\sigma^\mu p_\mu)_{\alpha\dot\beta}\,(n_B - n_F), \tag{2.5.9}$$

where n_B and n_F are the numbers of bosonic and fermionic states in V_p. For $p \neq 0$ the matrix $\sigma^\mu p_\mu$ is nonzero, so

$$\boxed{n_B = n_F.} \tag{2.5.10}$$

This is why the "ledger of degrees of freedom" of Chapter 0 has to balance.

2.5.3 Energy Can Never Be Negative

The last property is the most startling. By Eq. (2.2.30) from the example, for any state $|\psi\rangle$,

$$\langle\psi|H|\psi\rangle = \frac{1}{4}\sum_\alpha \left(\left\|Q_\alpha^\dagger|\psi\rangle\right\|^2 + \left\|Q_\alpha|\psi\rangle\right\|^2\right) \geq 0. \tag{2.5.11}$$

In a supersymmetric theory, no state can have negative energy. With $\mathcal{N}$ copies of the supercharges, you can do the same thing with any one copy, and the conclusion is unchanged.

Notice how much this statement says. In ordinary quantum field theory you can add any constant you like to the Hamiltonian, and the zero of energy has no meaning. In a supersymmetric theory you can't: H is completely determined by Eq. (2.2.30), with no room for an arbitrary constant. Equality holds if and only if every Q_α and $Q_\alpha^\dagger$ sends $|\psi\rangle$ to zero, that is, if the state is invariant under supersymmetry transformations. Applying this to the vacuum:

- If the vacuum is supersymmetric ($Q_\alpha|0\rangle = Q_\alpha^\dagger|0\rangle = 0$), the vacuum energy is exactly zero;
- Conversely, if the vacuum energy is positive, the vacuum cannot be annihilated by all the supercharges, and supersymmetry is **spontaneously broken**.

So the vacuum energy becomes an "order parameter" that tests whether supersymmetry is broken. We know supersymmetry must be broken in nature; this result tells us that the question comes down to: is the vacuum energy positive?

Example 2.2 Two special frames

Write $\{Q_\alpha, \bar{Q}_{\dot\beta}\}$ as a 2×2 matrix (a) in the rest frame of a massive particle, $p^\mu = (m,0,0,0)$, and (b) for a massless particle moving along z, $p^\mu = (E,0,0,E)$.

Solution (a) $p_\mu = (m,0,0,0)$, $\sigma^\mu p_\mu = m\,\mathbf{1}$, so $\{Q_\alpha, \bar{Q}_{\dot\beta}\} = 2m\,\delta_{\alpha\dot\beta}$. Regard $Q_\alpha/\sqrt{2m}$ and their conjugates as two pairs of fermionic creation and annihilation operators: they satisfy exactly the anticommutation relations of fermionic oscillators. (b) $p_\mu = (E,0,0,-E)$, $\sigma^\mu p_\mu = E(\mathbf{1}-\sigma_3) = \mathrm{diag}(0,2E)$, so

$$\{Q_1, \bar{Q}_{\dot1}\} = 0, \qquad \{Q_2, \bar{Q}_{\dot2}\} = 4E, \qquad \{Q_1, \bar{Q}_{\dot2}\} = \{Q_2, \bar{Q}_{\dot1}\} = 0.$$

By the norm argument in the text, $\{Q_1, Q_1^\dagger\} = 0$ means Q_1 vanishes identically on these states, and only Q_2 really does anything. Two pairs of "oscillators" in the massive case, only one pair in the massless case: this difference determines the sizes of massive and massless supermultiplets. □

***Problem 2.1** Expand the third line of Eq. (2.1.6) completely and verify that it holds for arbitrary operators F_1, F_2, B.

Problem 2.2 Suppose $[L^{\mu\nu}, Q_\alpha] = -(\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta$. Use the Jacobi identity $[[L^a, L^b], Q] = [L^a, [L^b, Q]] - [L^b, [L^a, Q]]$ (where a, b abbreviate the index pairs $\mu\nu$) to show that the matrices σ^a must satisfy the same commutation relations as the L^a, $[\sigma^a, \sigma^b] = \mathrm{i} f^{ab}{}_c \sigma^c$. What happens if we reverse the sign of Eq. (2.2.8), that is, $[L^{\mu\nu}, Q_\alpha] = +(\sigma^{\mu\nu})_\alpha{}^\beta Q_\beta$?

Problem 2.3 Take the Hermitian conjugate of Eq. (2.2.12) and show that $[\bar{Q}^{\dot\alpha}, P_\mu] = c^*(\bar\sigma_\mu)^{\dot\alpha\alpha} Q_\alpha$. (Hint: $(Q_\alpha)^\dagger = \bar{Q}_{\dot\alpha}$, $(\bar{Q}^{\dot\alpha})^\dagger = Q^\alpha$; then use Eq. (0.5.10).)

Problem 2.4 For Eq. (2.2.23), verify that the Jacobi identities $[\{Q_\alpha, \bar{Q}_{\dot\beta}\}, P_\mu] + \cdots = 0$ and $[\{Q_\alpha, Q_\beta\}, \bar{Q}_{\dot\gamma}] + [\{Q_\beta, \bar{Q}_{\dot\gamma}\}, Q_\alpha] + [\{\bar{Q}_{\dot\gamma}, Q_\alpha\}, Q_\beta] = 0$ hold for any c''.

***Problem 2.5** Let the bosonic state $|B\rangle$ have R-charge r. (a) Show that $Q_\alpha|B\rangle$ has R-charge $r+1$ and $\bar{Q}_{\dot\alpha}|B\rangle$ has R-charge $r-1$. (b) Show that $Q_\alpha Q_\beta|B\rangle = \frac{1}{2}\varepsilon_{\alpha\beta}(QQ)|B\rangle$, where $QQ \equiv Q^\alpha Q_\alpha$, and find the R-charge of $(QQ)|B\rangle$. (Hint: imitate Problem 0.25.)

Problem 2.6 (a) Explain why Z^{IJ} in Eq. (2.4.2) must be antisymmetric in IJ. (b) How many independent complex central charges are there for $\mathcal{N} = 2$ and for $\mathcal{N} = 4$? (c) Use a Jacobi identity to show that $[Z^{IJ}, P_\mu] = 0$.

****Problem 2.7** (a) Prove from Eqs. (0.2.14) and (0.5.26) that $\sigma^\nu \bar\sigma^\mu = \eta^{\mu\nu} + 2\mathrm{i}\sigma^{\mu\nu}$. (b) Use this to complete the derivation of $[\tilde{W}^\mu, Q_\alpha] = -\frac{1}{2} P^\mu Q_\alpha$ in the text.

***Problem 2.8** (a) Suppose supersymmetry is unbroken, $Q_\alpha|0\rangle = \bar{Q}_{\dot\alpha}|0\rangle = 0$. Show that $\langle 0|H|0\rangle = 0$. (b) Let $|B\rangle$ be a one-particle state at rest with mass $m > 0$. Show that at least one of the $Q_\alpha|B\rangle$ or $Q_\alpha^\dagger|B\rangle$ is nonzero. What does this tell you?